

- 1** An experiment to determine the acceleration of free fall g is conducted by projecting a stone with speed u at an angle θ to the horizontal. The horizontal distance R travelled by the stone when it returns to the level of projection is measured. Air resistance is negligible.

- (a)** In determining the speed of the stone, a student defines speed as “distance travelled per second”.

Explain why this definition is physically incorrect.

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 [2]

- (b)** Show that R is given by the expression

$$R = \frac{2u^2 \sin \theta \cos \theta}{g}.$$

[2]

- (c)** The above expression can be written as

$$R = \frac{u^2 \sin 2\theta}{g}.$$

The experiment is conducted to obtain the maximum range R_0 .

Determine the value of θ to obtain R_0 .

$\theta = \dots\dots\dots^\circ$ [1]

- (d) The values of u and R_0 are 12.6 m s^{-1} and 16.3 m , with associated percentage uncertainties of 3% and 4% respectively.

Calculate the value of g , and present the answer together with its uncertainty.

$$g = \dots\dots\dots \pm \dots\dots\dots \text{ m s}^{-2} \quad [3]$$

2 A light helical spring is suspended vertically from a fixed point as shown in Fig. 2.1